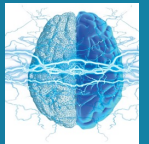


Quantifying the Effects of Cortical Fold Geometry on Electric Field Estimates in tCS Modeling

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Abstract

Transcranial current stimulation (tCS) relies on accurate 3D brain reconstructions to model current flow, but MRI-based reconstructions often distort fine cortical structures like sulci and gyri due to the partial volume effect. This project uses a spherical harmonics-based scaling method to systematically alter sulcal and gyral widths. By linking this geometric scaling to the electric field outcomes, this work characterizes the simulated relationship between cortical fold width and electric field estimates in 3D brain reconstructions for tCS modeling.

Methodology

- **Brain Surface:** MRI scans are used to generate a 3D cortical surface reconstruction in Freesurfer.
- **Cortical Scaling:** The brain surface is broken down into frequency components using spherical harmonics toolbox (SPHARM) in MATLAB. Specific components — called (L, M) pairs — were identified based on their influence on sulcal and gyral widths, then scaled by a range of factors to produce controlled geometric distortions mimicking cortical inaccuracies due to partial volume effects.
- **Mesh Generation:** The scaled cortical surface is combined with the remaining tissue surfaces (e.g., white matter, skull, scalp) to construct a tetrahedral finite element model using TetGen.
- **tCS Simulation:** For each modified brain geometry, tCS simulations are conducted to model electric field magnitude and spatial distribution. Electrode placement and stimulation parameters will be held constant across all conditions to isolate the effect of cortical geometry.

1. Picking test gyri

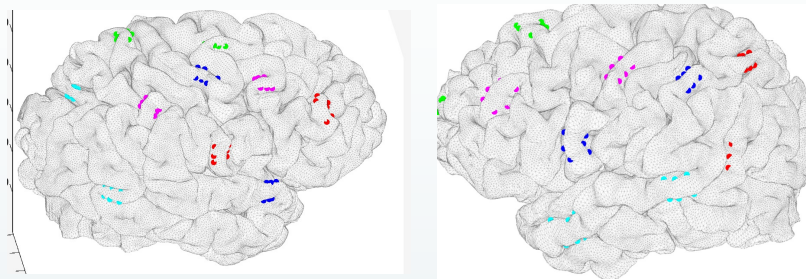


Fig 1. Chosen gyri for right and left hemispheres, these are the gyrus we will be manipulating.

2. L, M pair selection

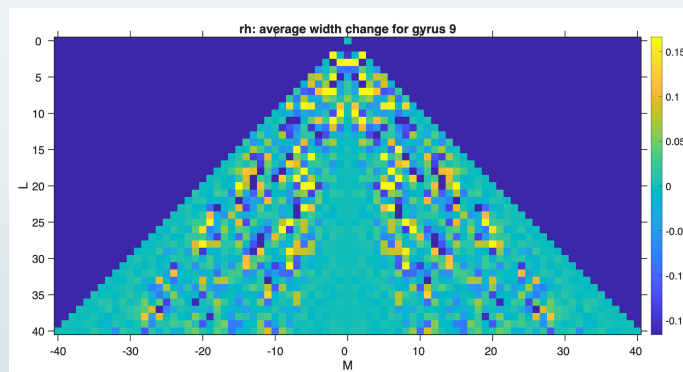


Fig 2. Heatmap for gyrus 9 from the right hemisphere, this tells us what L, M pairs have the biggest effect on the chosen gyrus from step 1.

3. Simulations

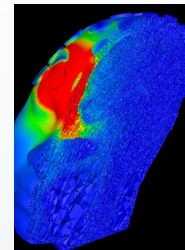


Fig 3. Example electric field inside the head from a tCS simulation in SCIRun. Study simulations are ongoing.

Conclusion

This work demonstrates that human brain surfaces can be accurately represented using spherical harmonics reconstructions. By analyzing SPHARM frequency components, we successfully identified the (L, M) pairs that govern sulcal and gyral width in a cortical surface. Systematically varying these coefficients produced valid brain surfaces with linearly increasing gyrus width, confirming the viability of this approach for generating controlled geometric distortions. These results establish a foundation for ongoing simulations that will quantify how such distortions affect modeled tCS electric fields.

Future Work

- Run tCS simulations on multiple models to help assess generalizability.
- Quantify a threshold of geometric distortion beyond which electric field estimates are significantly affected.